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Sustainable Biomaterials

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# RSB Greenhouse Gas Calculation Methodology

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# INTRODUCTION

This document details the methodology for the calculation of lifecycle greenhouse gas (GHG) emissions associated with RSB GLOBAL certified materials and fuels, biofuels, renewable liquid and gaseous transport fuels of non-biological origin (RFNBOs), recycled carbon fuels (RCFs), electrofuels (e-Fuels) and advanced products.
The methodology presented herein has been incorporated into the RSB Greenhouse Gas (GHG) Tool and RSB GHG Tool for RCF & RFNBO developed by RSB in partnership with Blue North Sustainability.

Main changes between Version 2.3 and Version 3.1:

a) Update of the RSB GLOBAL biofuel methodology.
b) Co-processing and flexible allocation of biofuels, RCFs and RFNBOs have been included.
c) The methodology for RCFs & RFNBOs production pathways are included.

Please see the full history of changes in Annex 1: History of changes of this Standard.

# LIST OF TABLES

*Table 1: Allocation approaches between products & co-products.* ..... 16
*Table 2: Fossil Fuel Comparators for Biofuels, RCFs, RFNBOs and e-Fuels.* ..... 16
*Table 3: Global Warming Potentials.* ..... 17
*Table 4: History of Changes to the RSB GHG Methodology documentation.* ..... 50
*Table 5: Biofuel and Biomaterial feedstock crops considered eligible under RSB GLOBAL.* ..... 51
*Table 6: List of Lower Heating Values.* ..... 52

# LIST OF ABBREVIATIONS

| Abbreviation | Definition |
|---|---|
| A | Area. |
| AB | Above-ground Biomass. |
| AF | Allocation Factor. |
| Aglime | Agricultural lime. |
| Art. | Article. |
| BB | Below-ground Biomass. |
| C | Carbon. |
| CF | Combustion Fraction. |
| $CH_4$ | Methane. |
| CHP | Combined Heat and Power Plant or cogeneration unit. |
| CoC | Chain of Custody. |
| $CO_2$ | Carbon Dioxide. |
| $CO_{2eq}$ | Carbon Dioxide equivalent. |
| CS | Carbon Stock. |
| D | Distance. |
| DOM | Dead Organic Matter. |
| DDGS | Distillers Dried Grains with Solubles. |
| E | Greenhouse Gas Emissions. |
| e-Fuels | Electrofuels. |
| EF | Emission Factor. |
| E.g. | For Example. |
| EU | European Union. |
| EU RED | European Renewable Energy Directive. |
| Eur | Euro. |
| Eq | Equivalence. |
| FF | Feedstock Factor. |
| GHG | Greenhouse Gas. |
| GWP | Global Warming Potential. |
| ha | Hectare. |
| ICAO | International Civil Aviation Organization. |
| I.e. | In other words. |
| IPCC | Intergovernmental Panel on Climate Change. |
| Kg | Kilogram. |
| L | Losses/Decrease. |
| LHV | Lower Heating Value. |
| LUC | Land Use Change. |
| M | Mass. |
| MC | Moisture Content. |
| N | Nitrogen. |
| N/A | Not Applicable. |
| $NH_3$ | Ammonia. |
| $NO_2$ | Nitrous Dioxide. |
| $NO_x$ | Nitrogen Oxides. |
| P | Productivity. |
| PoC | Proof of Sustainability. |
| RCF | Recycled Carbon Fuel. |
| RFNBO | Renewable Gaseous or Liquid Fuel of Non-Biological Origin. |
| RSB | Roundtable on Sustainable Biomaterials. |

| | |
|---|---|
| S&B | Stehfest and Bouwman. |
| SOC | Soil Organic Carbon. |
| STD | Standard. |
| Y | Yield. |

# TABLE OF CONTENTS


UNDERSTANDING THIS DOCUMENT ..... 10

**Aim of this document ..... 10**

**Scope of this document ..... 10**

**Status of this document ..... 11**

**Using this document ..... 11**

**References ..... 11**

**Terms & Definitions ..... 11**

REQUIREMENTS ..... 12

**A. General Requirements ..... 12**

**A.1. Functional Unit ..... 12**

**A.2. System Boundaries ..... 12**

**A.3. Geographic Scope ..... 12**

**A.4. Materiality Threshold Requirements ..... 13**

**A.5. Emission Factors ..... 13**

**A.6. Feedstocks ..... 13**

A.6.1. Eligible Biofuel & Biomaterial (Advanced Product) Feedstocks ..... 13

A.6.2. Eligible RCF & RFNBO Feedstocks ..... 14

*A.6.2.1. RCF Feedstocks ..... 14*

*A.6.2.2. RFNBO Feedstocks ..... 14*

**A.7. Moisture Content of Feedstock and Products ..... 14**

**A.8. Allocation Approaches between Products & Co-Products ..... 15**

**A.9. Greenhouse Gas Emission Savings ..... 16**

**A.10. Global Warming Potential ..... 17**

**B. Methodology for Biofuels ..... 18**

**B.1. Central Formula ..... 18**

**B.2. Feedstock & Allocation Factors ..... 18**

B.2.1. Feedstock Factors ..... 18

*B.2.1.1. Fuel & Energy supply chains ..... 18*

*B.2.1.2. Material supply chains ..... 18*

B.2.2. Allocation Factors ..... 18

B.2.3. Unit Conversion of Emissions ..... 18

B.2.4. Considering the Moisture Content of Feedstock and Products ..... 19

B.2.5. Chain of Custody ..... 19

**B.3. Emissions from Cultivation ..... 20**

B.3.1. Production-related Emissions of Seeding Material ..... 20

B.3.2. Production-related Emissions of Fertilisers and Pesticides ..... 21

*B.3.2.1. Emissions from Mineral Fertilisers ..... 21*

*B.3.2.2. Emissions from Organic Fertilisers ..... 21*

*B.3.2.3. Emissions from Plant Protection Products ..... 22*

*B.3.2.4. Emissions from Fertiliser Acidification and Liming Application ..... 22*

B.3.3. Emissions from Energy and Electricity-use for Cultivation-related Activities ..... 23

B.3.4. Soil (Nitrous Oxide) Emissions from Crop Cultivation ..... 23

*B.3.4.1. Direct Emissions ..... 24*

*B.3.4.2. Indirect Emissions ..... 24*

*B.3.4.3. Crop Residue Nitrogen input ($F_{CR}$) ..... 25*

*B.3.4.4. Crop and site-specific emission factors for $N_2O$ emissions from synthetic fertiliser and organic nitrogen application ..... 25*

*B.3.4.5. Emissions from Field Burning and Residue Removal ..... 26*

**B.4. Emissions from Land-Use Change (Annualised) ..... 27**

B.4.1. Change in Carbon Stocks in Biomass ..... 27

B.4.2. Change in Carbon Stocks in Dead Organic Matter ..... 28

B.4.3. Change in Carbon Stocks in Soils ..... 28

*B.4.3.1. Mineral Soils ..... 29*

*B.4.3.2. Organic Soils ..... 29*

**B.5. Emissions from Processing** ..... **30**

B.5.1. Combined Heat and Power Plants ..... 31

**B.6. Emissions from Transport and Distribution** ..... **32**

**B.7. Emissions of the Fuel in Use** ..... **33**

**B.8. Emission Savings from Improved Agriculture** ..... **33**

**B.9. Emission Allocation between Products & Co-Products** ..... **34**

**B.10. Co-Processing** ..... **34**

**B.11. Flexible Allocation** ..... **35**

**C. Methodology for Advanced Products** ..... **37**

**C.1. Central Formula** ..... **37**

**C.2. Emission Allocation between Products & Co-Products** ..... **37**

C.2.1. Economic Allocation ..... 37

**D. Methodology for RCFs, RFNBOs & e-Fuels** ..... **38**

**D.1. RCF, RFNBO & e-Fuel Fundamentals** ..... **38**

D.1.1. Chain of Custody ..... 38

D.1.2. Temporal Calculation ..... 39

**D.2. Central Formula** ..... **39**

**D.3. Emissions from the Supply of Inputs** ..... **40**

D.3.1. Elastic Inputs ..... 40

*D.3.1.1. Incorporated Processes ..... 41*

*D.3.1.2. Combined Heat and Power Plants ..... 41*

D.3.2. Rigid Inputs ..... 43

D.3.3. Inputs from Existing Use of Fate ..... 43

**D.4. Emissions from Processing** ..... **44**

**D.5. Emissions from Transport & Distribution** ..... **44**

**D.6. Emissions of Fuel in End Use** ..... **45**

**D.7. Emission savings from Carbon Capture and Geological Storage** ..... **45**

**D.8. Emission Allocation between Products & Co-Products ..... 45**

D.8.1. Energy Allocation ..... 46

D.8.2. Economic Allocation ..... 46

D.8.3. Two-Step Allocation Approach ..... 46

**D.9. Co-Processing ..... 47**

D.9.1. Share of RCF or RFNBO ..... 47

D.9.2. Relevant Energy Inputs ..... 47

*D.9.2.1. Material Inputs ..... 48*

*D.9.2.2. Electricity Inputs ..... 48*

*D.9.2.3. Heat Inputs ..... 48*

*D.9.2.4. Industrial Off-Gases ..... 48*

D.9.3. Emissions Intensity ..... 48

**D.10. Flexible Allocation ..... 49**

ANNEXES ..... 50

**Annex 1: History of changes ..... 50**

**Annex 2: Eligible Biofuel and Biomaterial Feedstocks ..... 51**

**Annex 3: Lower Heating Values ..... 52**

**Annex 4: References ..... 54**

# UNDERSTANDING THIS DOCUMENT

## Aim of this document

This standard aims to define the lifecycle GHG emissions calculation methodology that RSB-certified operators shall use for calculating and verifying GHG emissions and emissions reductions, ensuring compliance with the required minimum GHG savings compared to the fossil reference following RSB GLOBAL.

This methodology is embedded in the Biofuel and Biomass RSB GHG Tool and the RCF and RFNBO RSB GHG Tool for RSB GLOBAL Advanced Products and RSB GLOBAL Advanced Fuels.
This ensures consistent and standardized GHG emissions calculations across all certified entities whilst upholding chain of custody requirements.

## Scope of this document

This standard applies to all RSB-certified participating operators involved in the acquisition, handling, and/or forwarding of RSB GLOBAL certified materials, including biofuels, biomass fuels, renewable liquid and gaseous transport fuels of non-biological origin (RFNBOs), recycled carbon fuels (RCFs), e-Fuels, and advanced products.

While the RSB GHG Tools incorporate the RSB GLOBAL (global application), EU Renewable Energy Directive (EU RED; European Union application), and ICAO CORSIA (global application) lifecycle greenhouse gas emissions calculation methodologies, this document specifically details the RSB GLOBAL methodology. The methodologies detailed in this document include the emissions calculation related to RSB GLOBAL Advanced Fuels, or biofuels, in Chapter B, and RSB GLOBAL Advanced Products in Chapter C and RCFs and RFBNOs in Chapter D.

The RSB GLOBAL methodology is closely aligned with the EU RED. For the following product categories, RSB GLOBAL adheres to the methodologies detailed in the specified EU Directives and Regulations:

- **Biofuels:** *Annex V of Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018*.
- **Biomass Fuels:** *Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018, Annex VI*.
- **Extraction or cultivation of raw materials:** *Annex VII of Commission Implementing Regulation (EU) 2022/996 of 14 June 2022 alongside Intergovernmental Panel on Climate Change's (IPCC) 2019 Refinement of the 2006 Guidelines to National Greenhouse Gas Inventories Volume 4*.
- **RFNBO and RCF fuels:** *Commission Delegated Regulation (EU) 2023/1185 of 10 February 2023*.

For comprehensive details on the EU RED and ICAO CORSIA methodologies, please refer to the *RSB Standards and Procedures*[1] and to the official websites of the European Commission and ICAO, respectively. Where relevant, this document highlights key distinctions between the RSB GLOBAL methodology and these other frameworks; however, participating operators should consult the original sources for complete information.

## Status of this document

Version 3.0 of this RSB GHG Calculation Methodology shall be effective on 24th June 2025.
Whenever any contradiction or inconsistency exists between this version and previous versions of this standard, the latest version shall prevail. Any new version of this document will be notified immediately via email to all Participating Operators, Certification Bodies and RSB Accreditation Body.

## Using this document

All parts of this standard are normative, including its aim, scope, effective date, notes on its use, references, terms and definitions, requirements and annexes, unless stated otherwise. Participating operators implementing this standard shall ensure that all the requirements specified in this standard, and any other measures necessary to achieve its aim, are met. Participating operators of this standard should note that this document refers to greenhouse gas emissions expressed as carbon dioxide equivalents ($CO_{2eq}$) as "emissions".

## References

Please see *RSB List of Documents (RSB-DOC-10-001)* for the full list of RSB standards and references. Within this methodology document, refer to Annex 4: References for a list of all RSB and European Commission documents that are used within this RSB GLOBAL methodology document.

## Terms & Definitions

For the purposes of this document, the terms and definitions given in *RSB Glossary of Terms (RSB-STD-01-002)* shall apply.

Verbal forms for the expression of provisions include:

- **Shall** – Indicates requirements to be followed to conform to this document.
- **Should** – Indicates that among several possibilities, one is recommended as particularly suitable, without mentioning or excluding others, or that a certain course of action is preferred but not necessarily required.
- **May** – Indicates course of action permissible within the limits of the document.
- **Can** – Used for statements of possibility/capability, whether material, physical or causal.

[1] Refer to page 5 for documentation specific to each certification scheme: https://rsb.org/wp-content/uploads/2025/03/rsb-standards-and-certification-overview.pdf.

# REQUIREMENTS

## A. General Requirements

### A.1. Functional Unit

Emissions shall be expressed as follows:

1) For final biofuels / biomass fuels / RFNBO / RCF / e-Fuel, the functional unit shall be grams $CO_2$ equivalent per megajoule of final fuel product on a dry basis (at 0% water/moisture) [g $CO_{2eq}$/MJ fuel].
2) The megajoule content of fuel products must be determined using the Lower Heating Value (LHV) [kg/MJ fuel]. Refer to a full list of LHVs in Annex 3: Lower Heating Values.
3) For feedstock, some feedstocks and biomaterial-based products (advanced products), the functional unit shall be grams $CO_2$ equivalent per kilogram of feedstock or product on a dry basis (at 0% water/moisture) [g $CO_{2eq}$/kg feedstock or product].

### A.2. System Boundaries

For biofuels, the system boundary is from well (biofuel feedstock production) up to the use of the fuel in an engine. Theoretical (stoichiometric) emissions from fuel combustion shall thus be included. This approach is also called a well-to-wheel (road-based) or well-to-wake (aviation-based) analysis. For advanced products, the system boundary is from cradle (biomaterial feedstock production or extraction) to gate (the point where the product leaves the production facility gate) plus the distribution thereof. The use phase and end-of-life phase (e.g., disposal, recycling) shall be excluded.

For RCFs, RFNBOs, and e-Fuels, the system boundary encompasses the full life cycle emissions of the fuels, thereby from cradle-to-grave by considering the supply of inputs from the point of origin to the end-use combustion.

The boundary of the GHG calculation must coincide with the scope of certification. The boundary shall include all operations that are required to produce biofuel, biomaterial, RCF, RFNBO, or the processing of wastes. Some processing plants can be integrated with other operations; for example, an ethanol plant can be integrated with a combined heat and power plant (CHP). The operator must then deliver data on the whole operation.

### A.3. Geographic Scope

The RSB GLOBAL methodology is applicable to biofuel, biomaterial products, RCF, and RFNBO operations in any region of the world. Operators are required to apply operator-specific (i.e. actual) data (e.g., amount of fertiliser used) associated with their operations; hence, the methodology nor the RSB GHG Tools do not include default material and energy usage, nor assumptions on inventory data variability for different geographical areas, except for some cases where regional data is available such as electricity production.

## A.4. Materiality Threshold Requirements

All materials, chemicals and energy inputs into a system boundary must be considered in the calculations. The materiality threshold for inclusion in calculations is set at 1% of each category (material, chemical, and energy). Inputs below this threshold may be excluded. For instance, a chemical that amounts to less than 1% of all chemicals used within the system boundary is considered immaterial. Any inputs excluded from the GHG assessment due to the materiality thresholds shall be documented and justified and available for audits.

## A.5. Emission Factors

The emissions factors (EF) for the RSB GLOBAL methodology align with EU RED, thus, emissions from infrastructure and the manufacture of machinery and equipment (e.g., tractors, drilling equipment, refineries) shall not be included.

The emissions from processing the construction and decommissioning and waste management of electricity producing facilities are not considered. In addition, emissions from transport are based on specific fuel efficiencies, modes of transport and exhaust gas emissions. Emissions from transport and distribution shall include emissions from the storage and distribution of the finished fuels.

Waste and residues shall be considered to have zero emissions up to the process of collecting those materials.

The emissions associated with renewable electricity (wind, solar, hydro and geothermal) production are considered equal to zero.

In emission calculations according to this methodology under RSB GLOBAL, use the standard values of emission factors (*Annex IX of Commission implementing regulation (EU) 2022/996 of 14 June 2022*, or *Annex Part B and C of Commission delegated regulation (EU) 2023/1185 of 10 February 2023*). Where not possible, emission factors can be sourced from governmental publications of national emission factors. Alternatively, databases such as Ecoinvent or a similar, peer-reviewed, science-based life-cycle assessment databases or sources may be used.

## A.6. Feedstocks

### A.6.1. Eligible Biofuel & Biomaterial (Advanced Product) Feedstocks

Biofuel and biomaterial feedstocks considered under RSB GLOBAL shall include the following categories:

- Agricultural commodities (e.g., sugarcane).
- Agricultural and forestry residues (e.g., wheat straw).
- Processing residues (e.g., saw dust, bagasse, etc.).
- Animal residues from industrial processes (e.g., tallow).
- Waste oil.

For a full list of feedstock crops, refer to Annex 2: Eligible Biofuel and Biomaterial Feedstocks, in addition to *RSB Standard for Advanced Fuels (RSB-STD-01-010),* and *RSB EU RED Standard for Advanced Fuels (waste and residues) (RSB-STD-11-001-01-010).*

### A.6.2. Eligible RCF & RFNBO Feedstocks

#### A.6.2.1. RCF Feedstocks

RCFs are defined as fuels derived strictly from non-renewable waste or industrial byproducts that cannot be intentionally produced for fuel purposes. The eligibility of feedstocks for RCF production is guided by the Revised Directive (EU) 2018/2001 (also known as *Directive (EU) 2023/2413 of the European Parliament and of the Council of 18 October 2023 - RED III*) and the *Commission Delegated Regulation 2023/1185*. Two main RCF feedstock streams are allowed:

1. Non-renewable liquid or solid waste streams.

This includes waste materials that are unsuitable for material recovery as outlined by *Directive (EU) 2008/98/EC of 19 November 2008*. Producers must verify, through appropriate documentation, that these waste inputs cannot be feasibly recycled or reused, demonstrating that no viable recovery process is commercially available, and that the waste streams are of non-renewable origin.

2. Waste processing/exhaust gases.

This category covers non-renewable gases arising as incidental byproducts from industrial activities. These gases must be unavoidable emissions from primary production processes, such as exhaust gases, and cannot be produced intentionally for use in fuel production. Producers are required to provide documentation proving that these gases are either a waste or incidental outputs of an industrial process and would otherwise be released to the atmosphere.

#### A.6.2.2. RFNBO Feedstocks

The carbon, hydrogen and electricity which contribute to the energy content of the RFNBO shall have a non-biological origin. Eligible energy sources are from renewable sources other than biomass. The hydrogen component must be produced in an electrolyser that uses renewable electricity, excluding biomass and biogas electricity. In addition, the participating operator shall demonstrate that renewable hydrogen production via electrolysis complies with the *RSB Principles and Criteria (RSB-STD-01-001).*

Refer to *RSB Standard for Advanced Fuels (RSB-STD-01-010)* for feedstock specific requirements and specific requirements related to the production and use of renewable electricity.

## A.7. Moisture Content of Feedstock and Products

Based on the eligible feedstock, emissions per dry-kg feedstock or product shall be calculated as follows:
For feedstock i:

$$feedstock_i \left[\frac{gCO_2eq}{kg_{dry}}\right] = \frac{feedstock_i \left[\frac{gCO_2eq}{kg_{moist}}\right]}{(1 - MC_i)}$$

Where:
$MC_i$: Moisture content of feedstock i (fraction of water content within the total weight of the sample or feedstock)

## A.8. Allocation Approaches between Products & Co-Products

Where a fuel production process produces, in combination, the fuel for which emissions are being calculated and one or more other products (co-products), emissions shall be proportionally allocated between the fuel or its intermediate product and the co-products.

The methodology set out in this document shall follow an attributional analysis where environmental impacts are assigned or attributed to the biofuel (or advanced fuel) and its co-products according to a property, such as energy or economic value. The allocation reflects the relationship between the environmental burdens and the functions, i.e. how the burdens are changed by quantitative changes in the functions delivered by the system.

Consequential analysis (system expansion) involves conducting a full assessment of the fate of the co-product, what products in the market it will likely replace, and what effects this displacement will have on other markets. Attributional analysis through allocation allows for simpler analysis whilst considering factors that are under control of the operator. This approach eliminates the probability of results changing over time and the uncertainty associated with assumptions on product displacement and effects in international markets often made in consequential analysis. Therefore, credit allocation for trade or overall emission reductions in an offsetting scheme shall not be considered.

An intermediate product is the product of a specific stage within a larger, more complex production process that is further processed in sequential processes or stages to produce the system boundary's primary product.

Co-products are distinct from residues or wastes, which are defined as substances that are not the final products that a production process directly seeks to produce, and the process has not been deliberately modified to produce it. For a substance to be considered a co-product, continued use thereof is ensured with limited further pre-treatment. Examples of co-products are glycerin (or glycerol), biogas, or dried distillers' grains with solubles (DDGS). Examples of waste and residues are straw, husks, and crude glycerine.

For biofuels, all co-products are considered when allocating emissions. For RCFs and RFNBOs, refer to D.8. Emission Allocation between Products & Co-Products in this document that specifies the various approaches that can be applied for emissions allocation. Under RSB GLOBAL in accordance with EU RED, no emissions shall be allocated to waste, agricultural crop residues, or processing residues since they are considered to have zero emissions until the point of collection.

The following approaches shall be used to allocate emissions between the main/intermediate product and co-products (Table 1):

*Table 1: Allocation approaches between products & co-products.*

| Certification Scheme | System Type | Allocation Type |
|---|---|---|
| RSB GLOBAL Advanced Products | Advanced product | Economic |
| RSB GLOBAL Advanced Fuels | Biofuel | Economic |
| | RFNBO / RCF / e-Fuel | Two-Step Allocation Approach |

Energy allocation is performed when all co-products are fuels, electricity, or heat. In the case of energy allocation, emissions shall be divided in proportion based on the energy content (determined by lower heating value in the case of co-products other than electricity and heat). Co-products that have a negative energy content shall be considered to have an energy content of zero for the purposes of the calculation. Energy allocation shall be applied for RCFs, RFNBOs and e-fuels.

Economic allocation is based on the economic value of the co-products. It is useful when some co-products are materials with no energy content and reflects the value of the co-products in proportion to the main product. The economic value considered shall be the market value of the product and co-products as specified by the participating operator. Economic allocation shall be applied to RSB GLOBAL Advanced Products and RSB GLOBAL Advanced Fuels despite the energy nature of some of the products involved.

## A.9. Greenhouse Gas Emission Savings

Greenhouse gas emissions savings (percentage) from biofuels, RCFs, RFNBOs and e-fuels shall be calculated as follows:

$$Saving\ [\%] = \frac{(E_F - E)}{E_F} \times 100$$

Where:

E: Total emissions from the biofuel, RCF, e-Fuel, or RFNBO; and

$E_F$: Total emissions from the fossil fuel comparator.

For biofuels, RCFs, RFNBOs, and e-Fuels the fossil fuel comparator $E_F$ is indicated in the Table 2.

*Table 2: Fossil Fuel Comparators for Biofuels, RCFs, RFNBOs and e-Fuels.*

| Fossil comparator | RSB GLOBAL [g $CO_{2eq}$/MJ] | EU RED [g $CO_{2eq}$/MJ] | ICAO CORSIA [g $CO_{2eq}$/MJ] |
|---|---|---|---|
| Biofuel | 90 | 94 | 89 |
| RCF / RFNBO / e-Fuel | 94 | 94 | Yet to be defined by CORSIA |

Note:

- For GHG emission savings for biofuels (refer to section B. Methodology for Biofuels of this document), use the Biofuel and Biomass RSB GHG Tool.
- For GHG emission savings for RCFs and RFNBOs (refer to section D. Methodology for RCFs, RFNBOs of this document), use the RCF and RFNBO RSB GHG Tool.

- The fossil fuel comparator for advanced products shall be determined on a case-by-case study based on the actual lifecycle emissions value of the product it is replacing[2].

## A.10. Global Warming Potential

The GHGs considered shall be carbon dioxide ($CO_2$), nitrous oxide ($N_2O$) and methane ($CH_4$). For the purposes of calculating $CO_2$ equivalents ($CO_{2eq}$), the global warming potential (GWP) of those gases is defined in the (Table 3).

*Table 3: Global Warming Potentials.*

| Greenhouse Gas | RSB GLOBAL | EU RED | ICAO CORSIA |
|---|---|---|---|
| $CO_2$ | 1 | 1 | 1 |
| $N_2O$ | 265 | 265 | 265 |
| $CH_4$ | 28 | 28 | 28 |

The GHGs included in the calculation and their associated GWP for a 100-year basis align with the fifth assessment report (AR5) provided by the Intergovernmental Panel on Climate Change (IPCC) and the *Commission implementing regulation (EU) 2022/996*.

[2] Refer to *RSB Standard for Advanced Products (RSB-STD-02-001)* for specific criteria.

## B. Methodology for Biofuels

### B.1. Central Formula

Emissions from the production and use of biofuels shall be calculated as:

$$E\ \left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = e_{ec} + e_l + e_p + e_{td} + e_u - e_{sca} - e_{ccs} - e_{ccr}$$

Where:

E: Total emissions from the use of the biofuel.

$e_{ec}$: Emissions from the extraction or cultivation of raw materials.

$e_l$: Emissions (annualised) from carbon stock changes caused by land-use change.

$e_p$: Emissions from processing.

$e_{td}$: Emissions from transport and distribution.

$e_u$: Emissions from the fuel in use.

$e_{sca}$: Emission savings from soil carbon accumulation via improved agricultural management.

$e_{ccs}$: Emission savings from $CO_2$ capture and geological storage.

$e_{ccr}$: Emission savings from $CO_2$ capture and replacement.

### B.2. Feedstock & Allocation Factors

#### B.2.1. Feedstock Factors

Feedstock Factors (FFs) are defined as the ratio of feedstock required to make 1 unit of fuel of the system boundary's final product. This shall be determined on an energy basis (MJ ratio) or material basis (kg ratio).

##### B.2.1.1. Fuel & Energy supply chains

$$FF_{Energy_i} = [Ratio\ of\ MJ\ feedstock_i\ required\ to\ make\ 1\ MJ\ fuel]$$

##### B.2.1.2. Material supply chains

$$FF_{Material_i} = [Ratio\ of\ kg\ feedstock_i\ required\ to\ make\ 1\ kg\ biomaterial]$$

#### B.2.2. Allocation Factors

Allocation Factors (AFs) are defined as the fraction of the energy content of a product (or co-product) from the energy content of all products, including co-products, excluding wastes.

$$AF_i = \left[\frac{Energy\ in\ fuel_i[MJ_i]}{Energy\ in\ fuel_i[MJ_i] + Energy\ in\ all\ products[MJ_{all\ products}]}\right]$$

#### B.2.3. Unit Conversion of Emissions

Where the emissions for any of the terms in the central formula in Section B.1. must be converted from g $CO_{2eq}$/kg feedstock or product to g $CO_{2eq}$/MJ final fuel, the following calculation shall apply:

$$feedstock_i \left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = \frac{feedstock_i \left[\frac{g\ CO_{2eq}}{kg_{dry}}\right]}{LHV_i \left[\frac{MJ\ feedstock}{kg\ dry\ feedstock}\right]} \cdot FF_i \cdot AF_i$$

Refer to Annex 3: Lower Heating Values for a full list of LHVs.

#### B.2.4. Considering the Moisture Content of Feedstock and Products

The dry mass of a feedstock or product shall be calculated as follows:

$$dry\ mass\ of\ feedstock_i [kg_{dry}] = wet\ mass\ of\ feedstock_i [kg_{moist}] \times (1 - MC_i\ [\%])$$

Where:

MC: Moisture content (fraction on mass-basis).

Emissions per dry-kg feedstock or product shall be calculated as follows:

$$emissions\ of\ feedstock_i \left[\frac{g\ CO_{2eq}}{kg_{dry}}\right] = \frac{feedstock_i \left[\frac{g\ CO_{2eq}}{kg_{moist}}\right]}{(1 - MC_i\ [\%])}$$

Where:

MC: Moisture content (fraction on mass-basis).

The moisture content shall be the value measured upon delivery. If the moisture content is not known, the maximum value allowed per delivery contract shall be used.

#### B.2.5. Chain of Custody

The emissions of raw materials or intermediate products ($em_{cum(intermediate\ product)}$, as defined in the equation below) shall include the emissions of all inputs and raw materials, including the emissions of the previous step in the chain. Therefore, the feedstock factors (FF) and allocation factors (AF) of each supply chain step must be considered (as outlined in the equation below).

$$em_{cum\ intermediate\ product} \left[\frac{g\ CO_{2eq}}{kg_{dry}}\right] = Emissions_{\substack{actual \\ step}} \left[\frac{g\ CO_{2eq}}{kg_{dry}}\right] \cdot AF + Emissions_{\substack{previous \\ step}} \left[\frac{g\ CO_{2eq}}{kg_{dry}}\right] \cdot FF \cdot AF$$

Where:

$Emissions_{actual\ step}$: Emissions per kg product of the participating operator for the process under its scope of operation.

$Emissions_{previous\ step}$: Reported emissions per kg of product of the certified supplier, which include the emissions from previous steps up to cultivation (to be included if applicable to the specific system boundary).

## B.3. Emissions from Cultivation

Under RSB GLOBAL, the emissions from cultivation are aligned to EU RED in accordance with Annex VII of *Commission implementing regulation (EU) 2022/996* which follows the methodology set out by IPCC. Emissions from the extraction or primary production of raw materials, $e_{ec}$, shall include emissions from the extraction or cultivation process itself; from the collection, drying and storage of raw materials; from waste and leakages; and from the production of chemicals, fertilisers and other plant protection products used in extraction or cultivation. Carbon dioxide capturing (i.e. carbon removals) in the cultivation of raw materials shall be excluded.

The emissions from the extraction or cultivation process itself shall include (as presented in the equation below):

1) The production-related emissions of seeding material for crop cultivation ($E_{seed}$). Refer to B.3.1. Production-related Emissions of Seeding Material.
2) The production-related emissions of fertilisers ($E_{fertiliser}$) and plant protection products ($E_{pesticides}$). Refer to B.3.2. Production-related Emissions of Fertilisers and Pesticides.
3) The emissions from fertiliser acidification and liming application (included under $E_{fertiliser}$). Refer to B.3.2.4. Emissions from Fertiliser Acidification and Liming Application.
4) The emissions from the use of fuels and electricity in cultivation-related activities ($E_{fuel}$ and $E_{electricity}$). Refer to B.3.3. Emissions from Energy and Electricity-use for Cultivation-related Activities.
5) The soil (nitrous oxide) emissions from crop cultivation ($E_{Soil\ N2O}$). Refer to B.3.4. Soil (Nitrous Oxide) Emissions from Crop Cultivation.

$$e_{ec}\left[\frac{gCO_{2eq}}{ha}\right] = \frac{E_{seed}\left[\frac{gCO_{2eq}}{ha}\right] + E_{fertilizer}\left[\frac{gCO_{2eq}}{ha}\right] + E_{pesticides}\left[\frac{gCO_{2eq}}{ha}\right] + E_{fuel}\left[\frac{gCO_{2eq}}{ha}\right] + E_{electricity}\left[\frac{gCO_{2eq}}{ha}\right] + E_{Soil\ N_2O}\left[\frac{gCO_{2eq}}{ha}\right]}{product\ yield\left[\frac{kg}{ha}\right]}$$

Where:
Product yield shall be determined as the actual product realised in a particular harvest over the extent of the individual operation (production) site for crops that are harvested annually, or the actual product yield realised in the preceding 12 month period averaged over the extent of the individual operation (production) site for crops that are harvested more than once annually (or continually harvested).

### B.3.1. Production-related Emissions of Seeding Material

The emissions for production of seeding material shall be determined using the following equation based on the mass of seeds (specific to crop type) used per hectare. Please note, some of the terms used within the following sets of equations, are defined in other sections.

$$E_{seed_i}\left[\frac{g\ CO_{2eq}}{ha}\right] = mass\ of\ seed_i\ \left[\frac{kg}{ha}\right] \times EF_{production_{seed_i}}\left[\frac{g\ CO_{2eq}}{kg}\right]$$

Where:

$EF_{seed,i}$: Emission factor for the production-related emissions of seed type i.

### B.3.2. Production-related Emissions of Fertilisers and Pesticides

#### B.3.2.1. Emissions from Mineral Fertilisers

The production-related emissions for mineral fertilisers (nitrogen-, phosphorus, potassium-based, lime, etc.) shall be determined by considering the fraction of the active component within each type of fertiliser and the transportation of the mineral fertilisers. The emissions for mineral fertilisers are expressed through the equation below.

$$E_{fertilisers_{Mineral}}\left[\frac{g\ CO_{2eq}}{ha}\right] = \sum\left(m_{AI,fertiliser,i}\left[\frac{kg\ active\ ingredient}{ha}\right] \times EF_{AI,fertiliser,i}\left[\frac{g\ CO_{2eq}}{kg\ active\ ingredient}\right]\right) + \left(D_{type\ i}\ [km] \times \frac{m_{AI,fertiliser,i}\left[\frac{kg\ active\ ingredient}{ha}\right]}{f_{AI,fertiliser,i}\ [\%\ active\ ingredient]} \times EF_{Transport_i}\left[\frac{g\ CO_{2eq}}{kg \cdot km}\right]\right)$$

Where:

$D_{type,i}$: Transportation distance for mineral fertiliser type i.

$m_{AI,\ fertiliser,\ I}$: Mass of active ingredient/component type i applied per hectare.

$f_{AI,\ fertiliser,\ I}$: Fraction of active ingredient/component (e.g., N) in fertiliser product i.

$EF_{transport,I}$: Emission factor for transport type i.

$EF_{AI,fertiliser,I}$: Emission factor for mineral fertiliser application for active component i.

#### B.3.2.2. Emissions from Organic Fertilisers

Production-related emissions from application of organic fertilisers (such as solid manure from cattle, liquid pig manure, or vinasse) shall be determined by including the emissions from the transportation thereof from the point of production to application point. The emissions from organic fertilisers are expressed through the equation below.

$$E_{fertilisers_{Organic}}\left[\frac{g\ CO_{2eq}}{ha}\right] = \sum\left(D_{type\ i}[km] \times m_{fertiliser,i}\left[\frac{tonnes\ fertiliser}{ha}\right] \times EF_{Transport_i}\left[\frac{g\ CO_{2eq}}{tonnes \cdot km}\right]\right)$$

Where:

$D_{type,i}$: Transportation distance of organic fertiliser type i.

$m_{fertiliser,\ I}$: Mass of organic fertiliser i that is applied per hectare.

$EF_{transport,I}$: Emission factor for transport type i.

#### B.3.2.3. Emissions from Plant Protection Products

Production-related emissions from the application of plant protection products shall be determined by considering the percentage active component and total mass of pesticides, herbicides, insecticides or fungicides that are applied.

$$E_{pesticides}\left[\frac{g\ CO_{2eq}}{ha}\right] = \sum m_{pesticide,i}\left[\frac{kg}{ha}\right] \times f_{AI,pesticide,i}\ [\%\ active\ ingredient] \times EF_{pesticide,i}\left[\frac{g\ CO_{2eq}}{kg\ active\ ingredient}\right]$$

Where:

$m_{pesticide,\ I}$: Mass of plant protection product i applied per hectare.

$f_{AI,\ fertiliser,\ I}$: Fraction of active ingredient/component (e.g., urea compounds) in plant protection product type i.

$EF_{pesticide,I}$: Emission factor for plant protection product type i.

#### B.3.2.4. Emissions from Fertiliser Acidification and Liming Application

Emissions from fertiliser acidification and liming application are accounted for under the term; $E_{fertiliser}$, in the formula for emissions from cultivation (refer to Section B.3.1.).

Emissions from the neutralisation of fertiliser acidification and application of aglime (agricultural lime, a soil amendment used to neutralize soil acidity and improve soil health) shall account for the $CO_2$ emissions from neutralisation of acidity from nitrogen fertilisers (a) or from aglime reactions in the soil (b).

a) Under RSB GLOBAL in accordance with the Annex VII Part 1.4.2 of the *Commission implementing regulation (EU) 2022/996*, the default neutralisation emissions from nitrate fertilisers in the soil shall be 0.783 kg $CO_2$/kg Nitrogen (N), whereas for urea fertilisers, the neutralisation emissions shall be 0.806 kg $CO_2$/kg N.

b) The real amount of aglime used shall be duly documented. Emissions from aglime reactions in the soil shall be calculated based on the following approaches:

- On acidic soils, where the soil pH is less than 6.4, aglime is dissolved by soil acids to form predominantly $CO_2$ rather than bicarbonate, releasing almost all the $CO_2$ into the aglime (emission factor of 0.44 kg $CO_2$/kg $CaCO_3$eq aglime).
- If the soil pH is greater or equal to 6.4, an emission factor of 0.079 kg $CO_2$/kg $CaCO_3$eq aglime applied shall be considered in the calculation, in addition to the emissions due to the neutralisation of acidification caused by the fertiliser.

### B.3.3. Emissions from Energy and Electricity-use for Cultivation-related Activities

Emissions from the collection of raw materials include all the emissions resulting from the collection of raw materials and their transport to storage. The cultivation emissions shall include emissions from drying before storage, as well as from storage and handling of biomass feedstock.

Appropriate emission factors, including upstream emissions, shall be used to account for the emissions from the use of fuels (such as diesel oil, gasoline, heavy fuel oil, biofuels or other fuels) to produce heat or electricity used for drying or machinery use in other cultivation-related activities. Emissions for drying include only emissions for the drying process needed to ensure adequate storage of raw materials and do not include drying of materials during processing.

Emissions for fuel and electricity use in cultivation-related activities shall consider the sources and amounts of fuel and electricity used per hectare.

$$E_{fuel}\left[\frac{g\ CO_{2eq}}{ha}\right] = fuel_i\left[\frac{MJ}{ha}\right] \times EF_{fuel_i}\left[\frac{g\ CO_{2eq}}{MJ}\right]$$

$$E_{electricity}\left[\frac{g\ CO_{2eq}}{ha}\right] = electricity_i\left[\frac{kWh}{ha}\right] \times EF_{electricity_i}\left[\frac{g\ CO_{2eq}}{kWh}\right]$$

Where:

$EF_{Fuel,I}$: Emission factor of fuel source type i.

$EF_{Electricity,I}$: Emission factor of electricity source type i.

### B.3.4. Soil (Nitrous Oxide) Emissions from Crop Cultivation

Direct and indirect nitrous oxide emissions from managed soils shall follow the IPCC methodology with crop-specific emission factors for different environmental conditions corresponding to the Tier 2 approach[3].

$$N_{total} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right] = N_2O_{direct} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right] + N_2O_{indirect} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right]$$

Where:

$N_{total} - N$: Direct and indirect annual $N_2O$-N emissions produced from managed soils within the system boundary.

$N_2O_{direct} - N$: Annual direct $N_2O$-N emissions produced from managed soils.

$N_2O_{direct} - N$: Annual indirect $N_2O$-N emissions produced from atmospheric deposition of nitrogen volatilized from managed soils and annual amount of $N_2O$-N produced from leaching and run-off of nitrogen additions to managed soils in regions where leaching or run-off occurs.

[3] Tier 2 Approach (IPCC) is a method for estimating emissions that uses country-specific emission factors and more detailed activity data than the basic Tier 1.

#### B.3.4.1. Direct Emissions

For mineral soils:

$$N_2O_{Direct,mineral} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right] = [F_{SN} + F_{ON}] \times EF_{1,crop\ specific} + \left[F_{CR} \times EF_{1,crop\ specific}\right]$$

For organic soils:

$$N_2O_{Direct,organic} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right] = [F_{SN} + F_{ON}] \times EF_{1,crop\ specific} + \left[F_{CR} \times EF_{1,crop\ specific}\right] + \left[F_{OS,CG,Temp} \times EF_{2CG,Temp}\right] + \left[F_{OS,CG,Temp} \times EF_{2CG,Trop}\right]$$

#### B.3.4.2. Indirect Emissions

For mineral and organic soils:

$$N_2O_{indirect} - N\left[\frac{kg\ N_2O - N}{ha \cdot a}\right] = [\ [(F_{SN} \times Frac_{GASF}) + (F_{ON} \times Frac_{GASM}) \times EF_4] + \left[(F_{SN} + F_{ON} + F_{CR}) \times Frac_{Leach-(H)} \times EF_5\ \right]$$

Where:

$F_{SN}$: Annual synthetic nitrogen fertiliser input [kg synthetic N per ha per annum].

$F_{ON}$: Annual amount of animal manure, compost, sewage sludge and other organic nitrogen additions applied to soils [kg N per ha per annum].

$F_{CR}$: Annual amount of nitrogen in above and below ground crop residues [kg N per ha per annum].

$F_{OS,CG,Temp}$: Annual area of managed/drained organic soils under cropland in temperate climate [ha].

$F_{OS,CG,Trop}$: Annual area of managed/drained organic soils under cropland in tropical climate [ha].

$Frac_{GASF}$: Fraction of applied synthetic nitrogen fertiliser that volatilizes as $NH_3$ and $NO_x$ [kg N volatilized per kg of N applied].

$Frac_{GASM}$: Fraction of applied organic nitrogen fertiliser that volatilizes as $NH_3$ and $NO_x$ [kg N volatilized per kg of N applied or deposited].

$Frac_{Leach-(H)}$: Fraction of all nitrogen added to or mineralized in managed soils in regions where leaching/runoff occur [kg N per kg N additions].

$EF_{1ij}$: Crop and site-specific emission factors for $N_2O$ emissions from synthetic fertiliser and organic nitrogen application to mineral soils [kg $N_2O$-N per kg N input i].

$EF_1$: Emission factor for $N_2O$ emissions from nitrogen inputs to soils [kg $N_2O$-N per kg N input].

$EF_{2CG,\ Temp}$: Emission factor for temperate organic crops and grassland soils [kg $N_2O$-N per ha].

$EF_{2CG,\ Trop}$: Emission factor for tropical organic crops and grassland soils [kg $N_2O$-N per ha].

$EF_4$: Emission factor for $N_2O$ emissions from atmospheric deposition of nitrogen on soils and water surfaces [kg $N_2O$-N per kg $NH_3$-N + $NO_x$-N volatilized].

$EF_5$: Emission factor for $N_2O$ emissions from nitrogen leaching and runoff [kg $N_2O$-N per kg N leached or runoff].

#### B.3.4.3. Crop Residue Nitrogen input ($F_{CR}$)

a) For sugarcane, not considering below-ground residues and with the addition of N input from vinasse and filter cake in the case of sugarcane:

$$F_{CR,Sugarcane} = Y \cdot DRY \times \left(1 - Frac_{Burnt} \cdot C_f\right) \times [R_{AG} \cdot N_{AG} \cdot (1 - Frac_{Remove})] + F_{VF}$$

b) For coconut and oil palm plantations, a fixed N input is applied based on literature.[4]

c) For other crops:

$$F_{CR,Other\ crops} = \left(1 - Frac_{Burnt} \cdot C_f\right) \times AG_{DM} \cdot N_{AG} \cdot (1 - Frac_{Remove}) + (AG_{DM} + Y \cdot DRY) \times R_{BG-bio} \cdot N_{BG}$$

Where:

Y: Annual yield of the crop/product [kg/ha].

$Frac_{Burnt}$: Fraction of crop area burnt annually [ha burnt/ha crop].

$Frac_{Remove}$: Fraction of above-ground residues removed from field [kg dry matter per above ground dry matter].

$F_{VF}$: Annual amount of nitrogen in sugar cane vinasse and filter cake returned to the field [kg N/ha].

$AG_{DM}$: Above-ground residue dry matter [kg dry matter/ha].

$C_F$: Combustion factor [dimensionless].

$R_{AG}$: Ratio of above-ground residues to dry matter harvested yield for the crop [kg above ground residues per kg yield of dry matter].

$N_{AG}$: Nitrogen content of above-ground residues [kg N in above ground residues per kg dry matter].

DRY: Dry matter fraction of harvested crop/product [kg dry matter per kg fresh weight].

$N_{BG}$: Nitrogen content of below ground residues for crops [kg N in below ground residues per kg dry matter].

#### B.3.4.4. Crop and site-specific emission factors for $N_2O$ emissions from synthetic fertiliser and organic nitrogen application

$N_2O$ emissions from soils under agricultural use, in different agricultural fields under different environmental conditions and agricultural land use classes can be determined following the Stehfest and Bouwman (S&B model, 2006) statistical model, by following the approach outlined below:

[4] According to *Commission implementing regulation (EU) 2022/996*, 44 kg N per ha per annum for coconut; and 159 kg N per ha per annum for oil palm.

$$E_{N_2O}\left[\frac{kg\ N_2O-N}{ha\cdot a}\right]=\exp(-1.516+\Sigma ev)$$

Where:

$E_{N2O}$: Nitrous oxide emissions.

ev: Effect value for different drivers based on default values provided in Table 2 Annex VII of the *Commission implementing regulation (EU) 2022/996*.

Therefore, the $EF_{1ij}$ (crop and site-specific emission factors for $N_2O$ emissions from synthetic fertiliser and organic nitrogen application to mineral soils) for the biofuel crop i at location j is calculated with the S&B model below:

$$EF_{1ij}=\frac{E_{fert,ij}\left[\frac{kg\ N_2O-N}{ha\cdot a}\right]-E_{unfert,ij}\left[\frac{kg\ N_2O-N}{ha\cdot a}\right]}{N_{appl,ij}\left[\frac{kg\ N}{ha\cdot a}\right]}$$

The IPCC (2006) factor ($EF_1$) for direct $N_2O$ emissions from fertiliser input based on global averages shall be replaced by the crop- and site-specific $EF_{1ij}$ for direct emissions from mineral fertiliser and manure N input, based on the crop- and site- specific $EF_{1ij}$, applying the S&B model.

Where:

$E_{fert,ij}$: $N_2O$ emissions based on S&B, where the fertiliser input is the actual nitrogen application rate (mineral fertiliser and manure) to the crop i at location j.

$E_{unfert,ij}$: $N_2O$ emissions of the crop i at location j based on S&B. The nitrogen application rate is set to zero, all the other parameters are kept the same.

$N_{appl,ij}$: Nitrogen input from mineral fertiliser and manure to the crop i at location j.

#### B.3.4.5. Emissions from Field Burning and Residue Removal

The emissions from crop cultivation shall include emissions from burning sugarcane (agricultural residues) before harvesting in accordance with *Chapter 2 in IPCC's 2019 Refinement to the Guidelines for National Greenhouse Gas Inventories, Volume 4*.

$$E_{burn}[g\ GHG_i]=Area\ [ha]\times M_B\left[\frac{kg}{ha}\right]\times C_f\times EF_B\left[\frac{g\ GHG_i}{kg\ dry\ matter\ burnt}\right]$$

To then determine:

$$E_{burn,e}\left[g\ CO_{2eq}\right]=E_{burn}[g\ GHG_i]\times GWP_{GHG,i}$$

Where:

$E_{burn}$: Emissions from crops (sugarcane) burning before harvesting.

A: Area that is burnt before harvesting.

$M_B$: Mass of dry biomass that is burnt.

$C_f$: Combustion factor (default value of 0.8 for sugarcane).

EF: Emission factor for the mass of dry matter that is burnt.

GWP: Global warming potential.

## B.4. Emissions from Land-Use Change (Annualised)

Under RSB GLOBAL, annualized emissions from carbon stock changes caused by land-use change (LUC), $e_l$, are determined in accordance with *IPCC's 2019 Refinement to the 2006 Guidelines for National Greenhouse Gas Inventories, Volume 4*.

$$e_l \left[\frac{g\ CO_{2eq}}{kg\ crop_{dry} \cdot a}\right] = 3.67 \times 1000\ \left[\frac{kg}{tonnes}\right] \times 1000 \left[\frac{g}{kg}\right] \times \frac{\Delta C_{LU_1} \left[\frac{tonnes\ C}{ha \cdot a}\right]}{P_{dry} \left[\frac{kg\ crop_{dry}}{ha}\right]}$$

Where:

$e_l$: Annualized greenhouse gas emissions from carbon stock change due to land-use change.

$\Delta C_{LU1}$: Annual tonnes of carbon emitted per hectare, converted to tonnes carbon dioxide through the factor 3.67 derived from the molecular weight of carbon dioxide divided by the molecular weight of carbon.

$P_{dry}$: Productivity of the crop, measured as dry annual biomass harvested per hectare.

Estimating carbon stock changes and associated emissions of carbon dioxide from biomass, dead organic matter, and soils due to LUC (both land remaining in a land-use category as well as land converted to another land use) are determined by following the subsequent set of equations. The annual carbon stock changes (ΔC) for land-use category as a sum of changes in all pools shall be determined from the following equation.

$$\Delta C_{LU_1} \left[\frac{tonnes\ C}{ha \cdot a}\right] = \Delta C_{AB} + \Delta C_{BB} + \Delta C_{DOM} + \Delta C_{SO}$$

Where:

$\Delta C_{AB}$: Carbon stock changes in above ground biomass.

$\Delta C_{BB}$: Carbon stock changes in below ground biomass (default value is 0 under Tier 1 approach).

$\Delta C_{DOM}$: Carbon stock changes in dead organic matter.

$\Delta C_{SO}$: Carbon stock changes in soils (mineral and organic).

The carbon stock changes in each pool shall be determined from the following set of equations.

### B.4.1. Change in Carbon Stocks in Biomass

The sum of above-ground (AB) and below-ground biomass (BB) is the annual change in carbon stocks in biomass (B).

$$\Delta C_B \left[\frac{tonnes\ C}{ha \cdot a}\right] = \Delta C_{AB} + \Delta C_{BB}$$
$$= \Delta C_G \left[\frac{tonnes\ C}{ha \cdot a}\right]$$
$$+ \left(\left(0 - B_{BEFORE_i}\left[\frac{tonnes\ C}{ha \cdot a}\right]\right) \cdot \Delta A_{To\ Other}\ [ha]\right) \times CF\ \left[\frac{tonnes\ C}{tonnes\ dry\ matter}\right]$$
$$- \Delta C_L \left[\frac{tonnes\ C}{ha \cdot a}\right]$$

Where:

$\Delta C_G$: Annual increase in carbon stocks due to biomass growth.

$\Delta C_B$: Biomass stocks on land type i before the land-use conversion.

CF: Carbon fraction of dry matter.

$\Delta C_L$: Annual decrease in carbon stocks due to biomass loss.

A: Area of land use type i converted to another land-use category in a certain year.

### B.4.2. Change in Carbon Stocks in Dead Organic Matter

The annual change in carbon stocks in dead organic matter (DOM) due to land conversion shall be determined from the following equation:

$$\Delta C_{DOM} \left[\frac{tonnes\ C}{ha \cdot a}\right] = \frac{(C_n - C_o)\left[\frac{tonnes\ C}{tonnes\ dry\ matter \cdot ha}\right] \times \frac{A_{on}\ [ha]}{1\ [ha]}}{T_{on}\ [a]}$$

Where:

$C_n$: Deadwood/litter stock under the new land use category.

$C_o$: Deadwood/litter stock under the old land use category.

$A_{on}$: Area undergoing conversion from old to new land use category, divided by 1 hectare to convert the carbon stock change in DOM to per hectare-basis.

$T_{on}$: Time period of the land use transition (default is 20 years for carbon stock increases and 1 year for carbon losses).

### B.4.3. Change in Carbon Stocks in Soils

The change in carbon stocks in soils shall consider both mineral and organic soils. Organic soils have a minimum of 12-20% organic matter by mass. All other soils are classified as mineral soil types and typically have relatively low amounts of organic matter.

The effects of land-use and management activities on soil inorganic carbon stocks are linked to site hydrology and depend on the specific mineralogy of the soil and are therefore excluded from the calculations under Tier 1 and 2 approaches according to IPCC.

$$\Delta C_{SO} \left[\frac{tonnes\ C}{ha \cdot a}\right] = \Delta C_{Mineral} \left[\frac{tonnes\ C}{ha \cdot a}\right] - L_{Organic} \left[\frac{tonnes\ C}{ha \cdot a}\right]$$

Where:

$C_{Mineral}$: Annual change in organic carbon stocks in mineral soils (computed to a default depth of 30 cm under Tier 1 and 2).

$L_{Organic}$: Annual loss of carbon from drained organic soils.

#### B.4.3.1. Mineral Soils

The annual change in carbon stocks in mineral soils shall be determined based on the carbon stock after the management change (actual land use: 0-T) relative to the carbon stock in reference condition (reference land use: 0).

$$\Delta C_{Mineral}\left[\frac{tonnes\ C}{ha \cdot a}\right] = \frac{(SOC_0 - SOC_{0-T})\left[\frac{tonnes\ C}{ha}\right]}{D\ [a]}$$

Where:

SOC: Soil organic carbon.

D: Time dependence of stock change factors which is the default period for transition between equilibrium SOC values (default is 20 years).

Mineral soil carbon stocks per unit area of reference land-use and actual land-use shall be determined through the following equation:

$$SOC_{Mineral}\left[\frac{tonnes\ C}{ha}\right] = \Sigma_i\left(SOC_{Ref_i} \cdot F_{LU_i} \cdot F_{MG_I} \cdot F_{I_i} \cdot A_i\right)$$

Where:

$SOC_{Ref}$: The reference carbon stock.

$F_{LU}$: Stock change factor for land-use systems or sub-system for a particular land-use [dimensionless].

$F_{MG}$: Stock change factor for management regime [dimensionless].

$F_I$: Stock change factor for input of organic matter [dimensionless].

A: Land area [hectare] of the stratum being estimated.

#### B.4.3.2. Organic Soils

The annual carbon loss from drained organic soils shall be determined by assigning an annual emission factor that estimates the losses of carbon following drainage.

$$L_{Organic}\left[\frac{tonnes\ C}{ha \cdot a}\right] = \Sigma\left(\frac{A_c[ha]}{1[ha]} \cdot EF\left[\frac{tonnes\ C}{ha \cdot a}\right]\right)_c$$

Where:

$A_c$: Land area of drained organic soils in the specified climate type c, divided by 1 hectare to convert the annual carbon loss from drained organic soils to per hectare-basis.

$EF_c$: Emission factor for the specified climate type c.

## B.5. Emissions from Processing

Participating operators have the option to use actual values to cover all relevant processing steps. They can determine their actual processing emissions through calculations. If participating operators calculate actual values for processing, all relevant processing steps shall be considered. Emissions from processing, $e_p$, shall include emissions from processing itself (e.g., methane emissions from wastewater ponds); from waste and leakages; from the production of chemicals or products used in processing, and from energy and fuel use for processing operations.

Participating operators also have the option to use default values as set out by the European Commission for specific production pathways

In accounting for the consumption of electricity that is not produced within the system boundary (or fuel production plant), the greenhouse gas emissions intensity of the production and distribution of that electricity shall be assumed to be equal to the average emission intensity of the production and distribution of electricity in a defined region (geographical area in which fuel production occurs and where the electricity source is used). By way of derogation from this rule, producers may use an average value – for emission intensity - for an individual electricity production plant for electricity produced by that plant, if that plant is not connected to the electricity grid.

$$E_{fuel}\left[\frac{g\ CO_{2eq}}{kg}\right] = amount\ of\ fuel_i\ \left[\frac{MJ}{kg}\right] \times EF_{fuel_i}\left[\frac{g\ CO_{2eq}}{MJ}\right]$$

$$E_{electricity}\left[\frac{g\ CO_{2eq}}{kg}\right] = amount\ of\ electricity_i\ \left[\frac{kWh}{kg}\right] \times EF_{electricity_i}\left[\frac{g\ CO_{2eq}}{kWh}\right]$$

$$E_{Material}\left[\frac{g\ CO_{2eq}}{kg}\right] = \sum mass_{material}\left[\frac{kg}{kg\ product}\right] \times EF_{material}\left[\frac{g\ CO_{2eq}}{kg}\right]$$

Where:

$EF_{Fuel,I}$: Emission factor of fuel source type i.

$EF_{Electricity,I}$: Emission factor of electricity source type i.

$EF_{MaterialI}$: Emission factor of material, or chemical, or effluent source type i used for processing.

In the case of accounting for fuels produced in refineries, other than the combination of processing plants with boilers or cogeneration (CHP) units providing heat and/or electricity to the processing plant, the unit of analysis shall be the refinery.

### B.5.1. Combined Heat and Power Plants

Greenhouse gas emissions shall be calculated with the extension necessary for including the energy conversion to electricity and/or heat and cooling produced as follows:

i) For the electricity or mechanical energy coming from energy installations delivering useful heat together with electricity and/or mechanical energy:

$$EC_{el} = \frac{E}{\eta_{el}}\left(\frac{C_{el} \cdot \eta_{el}}{C_{el} \cdot \eta_{el} + C_h \cdot \eta_h}\right)$$

ii) For the useful heat coming from energy installations delivering heat together with electricity and/or mechanical energy:

$$EC_h = \frac{E}{\eta_h}\left(\frac{C_h \cdot \eta_h}{C_{el} \cdot \eta_{el} + C_h \cdot \eta_h}\right)$$

Where:

$EC_{el,h}$ : Total greenhouse gas emissions from the final energy commodity, electricity (el) or heat (h).

$E$: Total greenhouse gas emissions of the fuel before end-conversion.

$\eta_{el}$: The electrical efficiency, defined as the annual useful electricity produced by the annual fuel input based on its energy content.

$\eta_h$: The heat efficiency, defined as the annual useful heat output divided by the annual fuel input based on its energy content.

$C_{el}$: Fraction of exergy in the electricity, and/or mechanical energy, set to 100%.

$C_h$: Carnot efficiency (fraction of exergy in the useful heat).

Where a cogeneration unit (providing heat and/or electricity to a fuel production process for which emissions are being calculated) produces excess electricity and/or excess useful heat, the greenhouse gas emissions shall be divided between the electricity and the useful heat according to the temperature of the heat (which reflects the usefulness or utility of the heat). The useful part of the heat is found by multiplying its energy content with the Carnot efficiency, $C_h$, calculated as follows:

$$C_h = \frac{T_h - T_0}{T_h}$$

Where:

$T_h$: Temperature, measured in absolute temperature (kelvin) of the useful heat at point of delivery.

$T_0$: Temperature of surroundings, set at 273.15 kelvin (equal to 0 °C).

If the excess heat is exported for heating buildings, at a temperature below 150 °C (423.15 kelvin), $C_h$ can alternatively be defined as follows:

$C_h$ = Carnot efficiency in heat at 150 °C (423.15 kelvin), which is: 0.3546.

For the purposes of that calculation, the actual efficiencies shall be used, defined as the annual mechanical energy, electricity and heat produced respectively divided by the annual energy input. The following definitions apply:

i. "**Cogeneration**" shall mean the simultaneous generation in one process of thermal energy and electrical and/or mechanical energy.
ii. "**Useful heat"** shall mean heat generated to satisfy an economical justifiable demand for heat, for heating or cooling purposes.
iii. "**Economically justifiable demand**" shall mean the demand that does not exceed the needs for heat or cooling, and which would otherwise be satisfied at market conditions.

The GHG intensity of excess useful heat or excess electricity is the same as the greenhouse gas intensity of heat or electricity delivered to the fuel production process and is determined from calculating the greenhouse intensity of all inputs and emissions, including the feedstock and $CH_4$ and $N_2O$ emissions, to and from the CHP unit, boiler or other apparatus delivering heat or electricity to the fuel production process.

## B.6. Emissions from Transport and Distribution

If transportation is part of the operator scope, all relevant transport steps shall be considered using actual data. Emissions from transport and distribution, $e_{td}$, shall include emissions from the transport, storage, and filling of raw and semi-finished materials and from the storage and distribution of finished materials.

Transport may be divided into several separate transport steps. The emissions for transport shall be calculated for each transport step and summating them by using the equation outlined below.

$$E_{TD_i}\left[\frac{g\ CO_{2eq}}{kg}\right] = (D_{i,n}\ [km] \times L_i\ [\%] \times EF_i\ \left[\frac{g\ CO_{2eq}}{kg \cdot km}\right]) + \mathrm{E}_{\mathrm{Storage}}\ \left[\frac{g\ CO_{2eq}}{kg}\right]$$

Where:

$D_i$: Distance [km] travelled by the specific/selected transportation mode for the specified transport step & weight of material.

$L_i$: Percentage losses (mass-based) occurring during this transport/distribution step (i).

$EF_i$: Emission factor for the specified transport mode type i.

$E_{Storage}$: Emissions from electricity and fuel-use for storage and/or filling operations.

The following set of equations shall be used to determine emissions from storage and/or filling operations if electricity, energy, or heat are used.

$$\mathrm{E}_{\mathrm{Storage}}\left[\frac{g\ CO_{2eq}}{kg}\right] = E_{Energy}\left[\frac{g\ CO_{2eq}}{kg}\right] + E_{Electricity}\left[\frac{g\ CO_{2eq}}{kg}\right]$$

For which:

$$E_{Energy}\left[\frac{g\ CO_{2eq}}{kg}\right] = energy_i\ \left[\frac{MJ}{kg}\right] \times EF_{Energy_i}\left[\frac{g\ CO_{2eq}}{MJ}\right]$$

$$E_{Electricity}\left[\frac{g\ CO_{2eq}}{kg}\right] = electricity_i\left[\frac{kWh}{kg}\right] \times EF_{Electricity_i}\left[\frac{g\ CO_{2eq}}{kWh}\right]$$

Where:

$EF_{Energy,I}$: Emission factor for energy source type i used in storage and/or filling operations.

$EF_{Electricity,I}$: Emission factor for electricity source type i used for storage and/or filling operations.

## B.7. Emissions of the Fuel in Use

Fuel producers at the last interface are required to calculate the total emissions from the use of the fuel, including emissions from the end use ($e_u$) stage, for all carbon-containing fuels, ensuring that all lifecycle emissions are comprehensively accounted for. RSB GLOBAL is aligned with EU RED, under Annex V of the *Revised Directive (EU) 2018/2001*, which considers the following GHGs; $CO_2$, $N_2O$, and $CH_4$. Emissions from the fuel in use will always be positive if the combustion process produces the afore-mentioned GHGs. However, these emissions are considered zero if the fuel does not produce GHGs during combustion. For simplification reasons, the operator may also set the emissions of the fuel in use at zero instead of accounting for the biogenic carbon content in the feedstock.

Whilst the Biofuel and Biomass RSB GHG Tool allows the participating operator to enter their own emissions values for the fuel in use (fuel combustion emissions), as such emissions are dependent on engine performance and efficiency.

## B.8. Emission Savings from Improved Agriculture

Emissions savings from improved agriculture management, $e_{sca}$, such as shifting to reduced or zero-tillage, improved crop or rotation, the use of cover crops, including crop residue management, and the use of organic soil improver (e.g., compost, manure fermentation digestate), shall be taken into account only if solid and verifiable evidence is provided that the soil carbon has increased.

Participating operators seeking to claim emission savings from soil carbon accumulation via improved agricultural management should use the following formula in line with EU RED, based on the methodology outlined in Annex V of *Commission Implementing Regulation EU 2022/996* to determine their actual values and Annex VI Emission savings from soil carbon accumulation (esca) from *RSB Standard for EU Market Access (RSB-STD-11-001).*

$$e_{sca}\left[\frac{g\ CO_{2eq}}{MJ}\right] = (CS_A - CS_R)\left[\frac{Mg\ C}{ha}\right] \times \frac{44}{12} \times 10^6 \times \frac{1}{n} \times \frac{1}{P}\left[\frac{MJ}{ha \cdot a}\right] - e_f\left[\frac{g\ CO_{2eq}}{MJ}\right]$$

Where:

$CS_A$: Mass of soil carbon stock per unit area associated with the actual crop management practice after at least 10 years of application.

$CS_R$: Mass of soil estimated carbon stock per unit area associated with the reference crop management practice.

n: Number of years that the cultivation of the crop is considered.

P: Productivity of the crop measured as MJ Biofuel per hectare per year.

$e_f$: Emissions from the increased fertilisers or herbicides use.

The calculation of $CS_A$ and $CS_R$ shall be based on measurements of soil carbon stocks on farm level.

## B.9. Emission Allocation between Products & Co-Products

For biofuels under RSB GLOBAL, allocating emissions between main product and co-products follow the economic allocation approach (as outlined in Section A.8).

Economic Allocation:

$$AF_{co-produc_{Economic}} = \frac{V_{co-product_i}[currency]}{\sum_{i=1}^{n} V_{products_{i,n}}[currency]}$$

Where:

AF: Economic-based allocation factor for co-product or product.

i: For each product and co-product i of a process.

n: Total number of products and co-products of a process.

$V_i$: Economic value.

Note: The economic value considered shall be the average factory-gate value of the products over the last three years. If such data is not available, the market value of the commodity prices, minus the cost of transport and storage, shall be used.

## B.10. Co-Processing

The requirements for co-processing under RSB GLOBAL are detailed in the *RSB Proactive Guidance on Co-Processing GHG Calculations Version 1.0 (RSB-GUI-PG-2025-02).* This is in accordance with the *Commission Delegated Regulation (EU) 2023/1640 of 5 June 2023* on the methodology to determine the share of biofuel and biogas for transport, produced from biomass being processed with fossil fuels in a common process or refinery.

Emissions calculations for co-processing follow the approach of a virtual split of the biogenic conversion pathway into a baseline scenario and a biogenic feedstock scenario.

- The "**Baseline scenario"** includes all the inputs and emissions for the process operating with only fossil feedstock.
- The "**Biogenic feedstock scenario**" includes the biogenic feedstock either as an additional input or as a replacement for a portion of the fossil feedstock.

The share of biofuel or bio-based portion resulting from co-processing shall be determined through one of the following methods. It shall be expressed as the percentage of the total mass of the fuel (final product).

i. **Mass or energy balance calculations:** To determine the bio-based proportion of the product as detailed in Section 2.4.3 of *RSB Standard for Advanced Fuels (RSB-STD-01-010).*

ii. **Radiocarbon method:** To determine the biogenic carbon content as detailed in Section 2.4.4 of *RSB Standard for Advanced Fuels (RSB-STD-01-010).*

Emissions are expressed through the following set of equations:

$$E_{Biofuel}\left[\frac{g\ CO_{2eq}}{MJ\ Biofuel}\right] = e_{Upstream} + e_{p_{Co-P}} + e_{td} + e_u$$

Where:

$e_{Upstream}$: Upstream emissions from biogenic feedstock production, including cultivation, land use and transport to the primary production process (or refinery).

$e_{td}$: Emissions from transport and distribution of the final biofuel.

$e_u$: Emissions from the combustion of the biofuel during its final use.

$e_{p_{Co-P}}$: Processing emissions, determined through:

$$e_{p_{Co-P}} = \frac{E_{total\ process} \times F_{Bio}}{F_{Bio} + F_{Fossil}} + e_{additional}$$

Where:

$E_{total\ process}$: Total emissions from the 'Baseline scenario' shared between fossil and biogenic inputs.

$F_{Bio}$: Energy fraction of the biogenic feedstock at the inlet.

$F_{Fossil}$: Energy fraction of the fossil feedstock at the inlet.

$e_{additional}$: Additional emissions from energy or reagents specifically required (e.g., extra hydrogen or heat) for the biogenic feedstock.

Note: Inputs that are used for internal energy production or other auxiliaries, which will not be present in the final product, shall not be considered in determining the share of biofuel or bio-based portion resulting from co-processing.

If the primary method applied is not based on radiocarbon testing, participating operators shall use radiocarbon testing as a regular way of verifying and controlling the correctness of the performance of their system and the results of the main method used[5].

## B.11. Flexible Allocation

Flexible allocation of emissions in biofuel production refers to the mechanism of allowing participating operators to proportionally attribute processing emissions among biogenic and non-biogenic feedstocks or inputs through energy-based allocation.

[5] Refer to *RSB Proactive Guidance on Co-Processing GHG Calculations (RSB-GUI-PG-2025-02)* and *RSB Proactive Guidance on 14C Tests (RSB-GUI-PG-2025-01)* for more information.

To apply this mechanism, emissions are attributed for the energy inputs and reagents required in the 'Biogenic feedstock scenario' directly to the biofuel. Whilst, refinery-wide or process-wide energy inputs are proportionally allocated between the bio-based and fossil-based feedstocks, based on energy allocation. The 'Baseline scenario' shall be used to measure the increase in emissions due to the introduction of the biogenic feedstock. Any increases in inputs shall be allocated to the biofuel.

A more detailed information can be found in Annex IV: Optional flexible attribution methodology, in *RSB Standard for Advanced Fuels (RSB-STD-01-010).*

## C. Methodology for Advanced Products

### C.1. Central Formula

Greenhouse gas emissions from the production and use of advanced products shall be calculated as:

$$E\left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = e_{ec} + e_l + e_p + e_{td} - e_{sca}$$

Where:

E: Total emissions from the use of the biomaterial.

$e_{ec}$: Emissions from the extraction or cultivation of raw materials.

$e_l$: Annualized emissions from carbon stock changes caused by land-use change.

$e_p$: Emissions from processing.

$e_{td}$: Emissions from transport and distribution.

$e_{sca}$: Emission savings from soil carbon accumulation via improved agricultural management.

Refer to B.3. Emissions from Cultivation, B.4. Emissions from Land-Use Change (Annualised), B.5. Emissions from Processing, B.6. Emissions from Transport and Distribution, and B.8. Emission Savings from Improved Agriculture in this document for the calculation methodology for advanced products[6].

### C.2. Emission Allocation between Products & Co-Products

For advanced products under RSB GLOBAL, allocating emissions between products and co-products follow the economic approach indicated in Section A.8.

#### C.2.1. Economic Allocation

$$AF_{co-produc_{Economic}} = \frac{V_{co-product_i}[currency]}{\sum_{i=1}^{n} V_{products_{i,n}}[currency]}$$

Where:

AF: Economic-based allocation factor for co-product or product.

i: For each product and co-product i of a process.

n: Total number of products and co-products of a process.

$V_i$: Economic value.

Note: The economic value considered shall be the average factory-gate value of the products over the last three years. If such data is not available, the market value of the commodity prices, minus the cost of transport and storage, shall be used.

[6] Refer to *RSB Standard for Advanced Products (RSB-STD-02-001)* for specific criteria.

## D. Methodology for RCFs, RFNBOs & e-Fuels

### D.1. RCF, RFNBO & e-Fuel Fundamentals

RCFs are "liquid and gaseous fuels that are produced from liquid or solid waste streams of non-renewable origin which are not suitable for material recovery in accordance with *Article 4 of Directive (EU) 2008/98/EC*, or from waste processing gas and exhaust gas of non-renewable origin which are produced as an unavoidable and unintentional consequence of the production process in industrial installations"[7]. While RCFs are not classified as renewable because their carbon comes from non-renewable sources, they still contribute to emissions reduction by reusing carbon that would otherwise be emitted into the atmosphere.[8]

RFNBOs are "liquid or gaseous fuels the energy content of which is derived from renewable sources other than biomass"[9] . A fundamental prerequisite for RFNBOs is that they are produced using electricity generated from renewable energy technologies like wind, solar, geothermal, or hydropower. Unlike biofuels and e-Fuels, RFNBO production cannot rely on energy derived from biogenic sources such as biomass or biogas; instead, it harnesses renewable electricity to produce synthetic fuels through chemical processes.

Electrofuels (or e-fuels) are "liquid or gaseous fuels and intermediate products that are produced from renewable hydrogen together with carbon monoxide or carbon dioxide or nitrogen. Carbon and nitrogen can have biological and non-biological origins. The energy content stems from the hydrogen molecule in the fuel. Here, hydrogen must be produced in an electrolyser that uses renewable electricity or through renewable fuels reforming. Hydrogen from fossil hydrocarbon reforming shall not be accepted to produce e-fuels."[10]

#### D.1.1. Chain of Custody

Emissions for RFNBOs and RCFs are calculated by considering all relevant emission sources across the lifecycle, including inputs, processing, transportation, and end-use. Before the fuel reaches its final use, the supply chain's last participating operator must account for fuel combustion emissions.

As a general principle, all participating operators that add emissions to the product along the supply chain, including production, collection, processing, storing, or trading, must be covered by RSB certification. Certification must encompass the whole CoC, meaning that the tracking of materials and fuels must be transparent and verifiable from production to final use. Proof of Sustainability (PoS) documents in compliance with the established traceability and CoC requirements are necessary. This system ensures accurate accounting of renewable and non-renewable inputs and outputs.

Note: When renewable electricity is sourced via the grid through power purchase agreements (PPA), the hydrogen producer must submit data from a reliable, independent source, such as a grid operator or energy

[7] Refer to *Revised Directive (EU) 2018/2001, Art. 2 (35).*
[8] Capturing of emissions from non-sustainable sources should only be considered as avoiding emissions until 2035. Refer to *RSB Standard for Advanced Fuels (RSB-STD-01-010).*
[9] Refer to Art. 2 (36) in the *Revised Directive (EU) 2018/2001*.
[10] Defined in *RSB Standard for Advanced Fuels (RSB-STD-01-010).*

regulator. If such verification is not possible, certification from an independent third party providing hourly electricity production data may be required in accordance with the recognized standards. The relevant certification authority can provide a list of approved standards.

#### D.1.2. Temporal Calculation

The emissions intensity for RFNBOs and RCFs may be determined as an average for a production period of up to one calendar month, with shorter periods also permitted. The participating operator shall comply with the following rules:

- **Temporal correlation requirements:** When using fully renewable electricity (as defined by *Revised Directive (EU) 2018/2001 (RED III*) to enhance the heating value of products and intermediate products, the time intervals for emissions calculations shall meet the temporal correlation standards. Until 31 December 2029, calculations must be done every month; starting in 2030, they must be done hourly.

- **Averaging GHG values:** Emission values calculated for shorter intervals can be averaged over up to one month, provided all individual time intervals meet the minimum 70% GHG savings threshold. Intervals that do not meet the 70% threshold cannot be included in the monthly average[11].

## D.2. Central Formula

Under RSB GLOBAL, according to the *Delegated Regulation (EU) of 2023/1185*, the total emissions (E) for the production and use of the RCF or RFNBO shall be determined using the following formula by considering all relevant emission sources across the lifecycle, including all inputs, processing, transportation, and end-use:

$$E\left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = e_i + e_p + e_{td} + e_u - e_{ccs}$$

For which:

$$e_i\left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = e_{i\ elastic} + e_{i\ rigid} - e_{ex-use}$$

Where:

| | |
|---|---|
| $E$: | Total emissions from the use of the fuel. |
| $e_i$: | Emissions from the supply of inputs. |
| $e_{i\ elastic}$: | Emissions from elastic inputs. |
| $e_{i\ rigid}$: | Emissions from rigid inputs. |
| $e_{ex-use}$: | Emissions from inputs' existing use or fate. |
| $e_p$: | Emissions from processing. |
| $e_{td}$: | Emissions from transport and distribution. |
| $e_u$: | Emissions from combusting the fuel in its end use. |
| $e_{ccs}$: | Emission savings from carbon capture and geological storage. |

[11] Refer to Delegated Regulation (EU) of 2023/1185.

Note: Refer to A.5. Emission Factors for information on emission exclusions.

The emissions intensity for RFNBOs or RCFs production shall be determined by dividing the total emissions of the process covering each element of the formula by the total amount of fuel stemming from the process.

## D.3. Emissions from the Supply of Inputs

When determining emissions from the supply of inputs, emissions shall be categorised as rigid inputs, elastic inputs, and emissions from the inputs' existing use or fate.

All types of elastic and rigid inputs, along with their respective emissions from production and supply, must be accounted for under $e_i$. This also includes emissions from auxiliary processing inputs that are not part of the fuel's chemical composition.

Examples include electricity used in processing, water, heat, and other similar inputs. For these additional processing inputs, emissions are calculated using the relevant emission factors, which account for upstream emissions from production and supply of the input. These factors may be sourced similarly to those used for calculating emissions from elastic inputs, as outlined below.

### D.3.1. Elastic Inputs

Elastic inputs are those whose supply can be increased to meet extra demand of the RCF or RFNBO output production. Emissions from these inputs can fluctuate with market dynamics or production choices.

$$e_{i_{elastic_i}}[g\ CO_{2eq}] = EC_{elastic_i}[MJ_i] \times EF_{elastic_i}\left[\frac{g\ CO_{2eq}}{MJ_i}\right]$$

Or:

$$e_{i_{electricity}}[g\ CO_{2eq}] = amount\ of\ electricity_i\ [kWh]\ \times\ EF_{electricity}\left[\frac{g\ CO_{2eq}}{kWh}\right]$$

Where:

$EC_{elastici}$: Energy content of the elastic input type i.

$EF_{Electricity,I}$: Emission factor of electricity source type i.

Under RSB GLOBAL, 'energy from renewable sources' or 'renewable energy' means energy from renewable non-fossil sources, namely wind, solar (solar thermal and solar photovoltaic) and geothermal energy, osmotic energy, ambient energy, tide, wave and other ocean energy, hydropower, biomass, landfill gas, sewage treatment plant gas, and biogas[12].

[12] As defined in *RSB Standard for Advanced Fuels (RSB-STD-01-010).*

Electricity qualified as fully renewable according to Article 27(6) of *Revised Directive (EU) 2018/2001 (RED III),* shall be attributed zero emissions. Electricity taken from the grid that does not qualify as fully renewable, and is used to produce RCFs and RFNBOs, follows the alternative approach outlined in Part A.6(a) Annex I of the *Commission Delegated Regulation (EU) 2023/1185* to attribute emission values. Under RSB GLOBAL, the participating operator shall comply with the rules stated in Part C Annex I of *Commission Delegated Regulation (EU) 2023/1185* for alternative approaches in determining the emission intensity of electricity.

#### D.3.1.1. Incorporated Processes

Incorporated processes include processes that take place in the same industrial complex, or that supply the input via a dedicated supply of infrastructure, or that supply more than half of the energy of all inputs to the production of RCFs and/or RFNBOs. Emissions of elastic inputs obtained from an incorporated process shall be included in the emissions calculations; these are determined based on data from their actual production process.

This shall include all emissions arising from their production over the whole supply chain (including emissions arising from the extraction of the primary energy required to make the input, processing of the input and transportation of the input). Combustion emissions related to the carbon content of fuel inputs shall not be included since it is accounted for in processing or the combustion of the final fuel.

$$e_{i_{elastic_{incorporated}}}[g\ CO_{2eq}] = \left(EC_{elastic_i}[MJ_i] \times EF_{elastic_i}\left[\frac{g\ CO_{2eq}}{MJ_i}\right]\right) + e_{incorporated}\ [g\ CO_{2eq}]$$

Where:

$e_{incorporated}$: Processing, transport and loss of production emissions for elastic inputs from incorporated processes.

If elastic inputs are not sourced from incorporated processes, emissions shall be determined based on the EF values provided in *Annex I from Commission Delegated Regulation EU/2023/1185* or sourced from a reputable emission factor database such as the Ecoinvent database.

#### D.3.1.2. Combined Heat and Power Plants

Electricity or heat from a CHP are elastic inputs. Emissions shall be calculated with the extension necessary for including the energy conversion to electricity and/or heat and cooling produced as follows:

i. For the electricity or mechanical energy coming from energy installations delivering useful heat together with electricity and/or mechanical energy:

$$EC_{el} = \frac{E}{\eta_{el}}\left(\frac{C_{el} \cdot \eta_{el}}{C_{el} \cdot \eta_{el} + C_h \cdot \eta_h}\right)$$

ii. For the useful heat coming from energy installations delivering heat together with electricity and/or mechanical energy:

$$EC_h = \frac{E}{\eta_h}\left(\frac{C_h \cdot \eta_h}{C_{el} \cdot \eta_{el} + C_h \cdot \eta_h}\right)$$

Where:

$EC_{el,h}$: Total emissions from the final energy commodity, electricity (el) or heat (eh).

$E$: Total emissions of the fuel before end-conversion.

$\eta_{el}$: The electrical efficiency, defined as the annual electricity produced divided by the annual fuel input based on its energy content.

$\eta_h$: The heat efficiency, defined as the annual useful heat output divided by the annual fuel input based on its energy content.

$C_{el}$: Fraction of exergy in the electricity, and/or mechanical energy, set to 100% ($C_{el} = 1$).

$C_h$: Carnot efficiency (fraction of exergy in the useful heat).

Where a CHP unit – providing heat and/or electricity to a fuel production process for which emissions are being calculated – produces excess electricity and/or excess useful heat, the emissions shall be divided between the excess electricity and/or the useful heat according to the temperature of the heat (which reflects the usefulness, or utility, of the heat). The useful part of the heat is found by multiplying its energy content with the Carnot efficiency, $C_h$, calculated as follows:

$$C_h = \frac{T_h - T_0}{T_h}$$

Where:

$T_h$: Temperature, measured in absolute temperature (kelvin) of the useful heat at point of delivery.

$T_0$: Temperature of surroundings, set at 273.15 kelvin (equal to 0 °C).

If the excess heat is exported for heating of buildings, at a temperature below 150 °C (423.15 kelvin), $C_h$ can alternatively be defined as follows:

$C_h$: Carnot efficiency in heat at 150 °C (423.15 kelvin), which is: 0.3546.

For the purposes of that calculation, the actual efficiencies shall be used, defined as the annual mechanical energy, electricity and heat produced respectively divided by the annual energy input. For the purposes of that calculation, the following definitions apply:

i. "**Cogeneration**" shall mean the simultaneous generation in one process of thermal energy and electrical and/or mechanical energy.
ii. "**Useful heat**" shall mean heat generated to satisfy an economical justifiable demand for heat, for heating or cooling purposes.
iii. "**Economically justifiable demand**" shall mean the demand that does not exceed the needs for heat or cooling, and which would otherwise be satisfied in market conditions.

The emissions from the excess useful heat or excess electricity are the same as the emissions intensity of heat or electricity delivered to the fuel production process and are determined from calculating the greenhouse intensity of all inputs and emissions, including the feedstock and $CH_4$ and $N_2O$ emissions, to and from the CHP unit, boiler or other apparatus delivering heat or electricity to the fuel production process.

### D.3.2. Rigid Inputs

Rigid inputs are those whose supply cannot be expanded to meet additional demand. All inputs qualifying as a carbon source to produce RCFs are thus considered rigid, as well as outputs produced in a fixed ratio within an incorporated process. If an output represents less than 10% of the economic value of the total output, it is categorised as rigid; however, if it represents 10% or more of the economic value, it is treated as elastic.

Emissions from rigid inputs must account for their supply, extraction, transportation, and shall include the emissions resulting from the diversion of those inputs from a previous or alternative use. These emissions shall consider the loss of production of electricity, heat or products that were previously generated using the input, as well as any emissions due to additional treatment of the input and transport. Refer to Annex I Part 9 in the *Commission Delegated Regulation (EU) 2023/1185* for alternative approaches in determining emissions attributed to the supply of rigid inputs.

$$e_{i_{rigid_i}} = e_{diversion}\ [g\ CO_{2eq}]$$

Where:

$e_{diversion}$: Emissions from loss of production, additional treatment of the input, and transport of the input.

### D.3.3. Inputs from Existing Use of Fate

Emissions from existing use or fate include all emissions that are avoided when using such input for fuel production. This includes $CO_2$ that was captured and incorporated into the fuel provided that at least one of the following conditions is fulfilled:

a) The $CO_2$ has been captured from an activity listed under Annex I of *Directive (EU) 2003/87 of the European Parliament and of the Council of 13 October 2003* as amended by *Directive (EU) 2023/959 of 10 May 2023*; and has been considered upstream in an effective carbon pricing system and is incorporated in the chemical composition of the fuel before 2036. This date shall be extended to 2041 in other cases than $CO_2$ stemming from the combustion of fuels for electricity generation. For $CO_2$ captured outside of the Member States, a national and robust system of effective carbon pricing must be in place; or

b) The $CO_2$ has been captured from the air; or

c) The captured $CO_2$ stems from the production or the combustion of biofuels, bioliquids or biomass fuels complying with the *RSB Principles & Criteria (RSB-STD-01-001)* and the $CO_2$ capture did not receive credits for emission savings from $CO_2$ capture and replacement, set out in Annex V and VI of the *Revised Directive (EU) 2018/2001 (RED III)*; or

d) The captured $CO_2$ stems from the combustion of RFNBO or RCF complying with the greenhouse gas saving criteria, set out in Article 29a of *Revised Directive (EU) 2018/2001 (RED III)* and in *Commission Delegated Regulation (EU) 2023/1185*; or

e) The captured $CO_2$ stems from a geological source of $CO_2$ and the $CO_2$ was previously released naturally.

Emissions associated with the inputs like electricity, heat, and consumable materials used in the capture process of $CO_2$ shall be included in the calculation of emissions attributed to inputs.

## D.4. Emissions from Processing

This covers the emissions generated during the processing of the inputs into RFNBOs or RCFs. These include emissions from electricity, heat, and other resources used in the conversion processes. Emissions from processing shall include direct atmospheric emissions from the processing itself, from waste treatment and from leakages. Such leakages can for example originate from the use of $CO_2$. Where the use of processing inputs results in direct atmospheric emissions, such as combustion, these emissions will then be accounted for under $e_p$.

$$E_p[g\ CO_{2eq}] = \sum m_{effluent_i}[kg\ or\ m^3] \times EF_{effluent_i} \left[\frac{g\ CO_{2eq}}{kg\ or\ m^3}\right]$$

Where:

$m_{effluent,i}$: The volume of wastewater treated [$m^3$], or mass of methane/nitrous oxide emissions from the process facility [kg].

## D.5. Emissions from Transport & Distribution

Emissions from transport and distribution shall include emissions from the storage and distribution of the finished fuels. Emissions attributed to inputs $e_i$ shall include emissions from their associated transport and storage.

$$E_{TD_{i,n}}[g\ CO_{2eq}] = (D_{i,n}\ [km] \times m_i[kg] \times L_{i,n}\ [\%] \times EF_{i,n} \left[\frac{g\ CO_{2eq}}{kg \cdot km}\right]) + \mathrm{E}_{\mathrm{Storage}}[g\ CO_{2eq}]$$

Where:

$D_i$: Distance travelled by the specific/selected transportation mode for the specified transport step & weight of material.

$m_i$: Mass of material that is transported in the specified transport mode i.

$L_i$: Percentage losses (mass-based) occurring during this transport/distribution step (i).

$EF_i$: Emission factor for the specified transport mode type i.

$E_{Storage}$: Emissions from electricity and fuel-use for storage and/or filling operations.

The following set of equations are used to determine emissions from filling and storage operations if electricity, energy, or heat are used in any of the processes.

$$\mathrm{E}_{\mathrm{Storage}}[g\ CO_{2eq}] = E_{Energy}[g\ CO_{2eq}] + E_{Electricity}[g\ CO_{2eq}]$$

For which:

$$E_{Energy}[g\ CO_{2eq}] = \sum amount\ of\ energy_i[MJ] \times EF_{Energy_i}\left[\frac{g\ CO_{2eq}}{MJ}\right]$$
$$E_{Electricity}[g\ CO_{2eq}] = \sum amount\ of\ electricity_i\ [kWh] \times EF_{Electricity_i}\left[\frac{g\ CO_{2eq}}{kWh}\right]$$

Where:

$EF_{Energy,I}$: Emission factor for energy source type i used for storage and/or filling operations.

$EF_{Electricity,I}$: Emission factor for electricity source type i used for storage and/or filling operations.

## D.6. Emissions of Fuel in End Use

Fuel producers at the last interface are required to calculate the total emissions from the use of the fuel, including emissions from the end-use ($e_u$) stage, for all carbon-containing fuels, ensuring that all lifecycle emissions are comprehensively accounted for. RSB GLOBAL is aligned with EU RED, under Annex V of the *Revised Directive (EU) 2018/2001*, which considers the following GHGs; $CO_2$, $N_2O$, and $CH_4$. Emissions from the fuel in use will always be positive if the combustion process produces the afore-mentioned GHGs. However, these emissions are considered zero if the fuel does not produce GHGs during combustion; this applies to fuels like hydrogen and ammonia. For simplification reasons, the operator may also set the emissions of the fuel in use at zero instead of calculating and deducting avoided emissions from the final GHG value.

## D.7. Emission savings from Carbon Capture and Geological Storage

Where a process for making RFNBO or RCF produces carbon emissions that are permanently stored in accordance with *Directive (EU) 2009/31/EC of 23 April 2009* on the geological storage of carbon dioxide, this may be credited to the products of the process as a reduction in emissions under $e_{ccs}$. Emissions arising due to the storage operation (including transport of the carbon dioxide gas) will also need to be considered under $e_p$.

## D.8. Emission Allocation between Products & Co-Products

When a process yields multiple co-products such as fuels or chemicals, as well as energy co-products such as heat, electricity or mechanical energy exported from the plant, emissions shall be allocated to these co-products applying the following approaches. The allocation shall be conducted at the end of the process that

produces the primary or intermediate product and co-products. Allocated emissions shall include the emissions from the process itself, as well as the emissions attributed to inputs to the process.

#### D.8.1. Energy Allocation

Where the ratio of the products is fixed and the co-products are all fuels, electricity or heat, the allocation shall be done by energy content. If allocation concerns exported heat based on the energy content, only the useful part of the heat may be considered as defined in Part C (Point 16) Annex V of *Directive (EU) 2018/2001*. This is the preferred method.

$$AF_{co-product_{Energy}} = \frac{C_{co-product\ _i}[MJ]}{\sum_{i=1}^{n} C_{products_{i,n}}[MJ]}$$

Where:

AF: Energy-based allocation factor for the co-product or product.

i: For each product i of a process, including co-products.

n: Total number of products of a process, including co-products.

$C_{output_i}$: Energy content of all products and co-products n in a process.

#### D.8.2. Economic Allocation

Where the ratio of products is fixed, however some of the co-products are materials with no energy content, the allocation shall be done by the economic value of the co-products. The economic value shall be the average factory-gate value (total market value) of the products over the last three years. If such data is not available, the value shall be estimated from commodity prices minus the cost of transport and storage.

$$AF_{co-produc_{Economic}} = \frac{V_{co-product_i}[Eur]}{\sum_{i=1}^{n} V_{products_{i,n}}[Eur]}$$

Where:

AF: Economic-based allocation factor for co-product or product.

i: For each product and co-product i of a process.

n: Total number of products and co-products of a process.

$V_i$: Economic value.

#### D.8.3. Two-Step Allocation Approach

When the ratio of products is fixed and a mix of energy and material co-products are produced, a two-step allocation method shall be applied to determine the allocation factors for all products:

a) Economic allocation is applied first to separate materials from fuels (refer to Section D.8.2. for the equation).

b) After deducting from the total, the emissions allocated to materials, energy content allocation is then applied to the energy-containing products (refer to Section D.8.1. for the equation).

Note: If any installation inside the project boundary treats only one of the project's co-products, then the emissions from that installation shall be ascribed entirely to that co-product. Where multiple sub-processes are involved, emissions are allocated only to relevant co-products.

## D.9. Co-Processing

If a fuel (main product produced within a system boundary) is a mix of RCF, RFNBO, and any other fuels, all fuel types shall be considered to have the same emission intensity [g $CO_{2eq}$/MJ Fuel]. The exception to this rule is the case of co-processing. When RCFs and RFNBOs are only partially replacing a conventional input in a process or are processed together with biomass, their (RCF or RFNBO) respective shares in the total output shall be determined by applying the concept of "relevant energy input". From which emissions intensity is determined on a proportional basis of the energetic value of inputs (refer to Section D.9.2.). The same approach applies if the output of a process does not fully qualify as RCF or RFNBO.

### D.9.1. Share of RCF or RFNBO

The respective share of RCF or RFNBO in the total output shall be determined using the following approaches.

- The share (S) of RCF shall be determined by dividing the relevant energy input qualifying as a source to produce RCF by the total relevant energy inputs into the process.
- The share (S) of RFNBO shall be determined by dividing the relevant renewable energy input into the process by the total relevant energy inputs.

$$S_{RFNBO/RCF,out} = \frac{E_{RFNBO/RCF,in}}{E_{educts}}$$

Where:

$S_{RFNBO/RCF,out}$: Share of the product that can be certified as RFNBO or RCF.

$E_{RFNBO/RCF,in}$: Energy content of the RFNBO or RCF inputs only.

$E_{educts}$: Energy content of all relevant inputs (defined in Section D.9.2.)

Note: Only relevant inputs are considered in the equation above to determine the volume of qualifying RCF or RFNBO that is produced. Other inputs ("non-relevant") are only considered when determining the emission intensity of the fuel, not for the fraction (or share) of RCF or RFNBO in the final product.

### D.9.2. Relevant Energy Inputs

The concept of "relevant energy input" refers to the energy contribution from material, electricity, heat, or industrial off-gases that directly enhances the heating value or molecular structure of the RCF or RFNBO fuel being produced.

Relevant energy inputs to the system shall be determined by the participating operator through comparison of the heating value of the inputs entering the process and the heating value of the RCF or RFNBO output.

The relevant energy inputs shall be determined using various methods, depending on the input type as defined below.

#### D.9.2.1. Material Inputs

For material inputs, the relevant energy input is the LHV of the dry part of the material input that enters the fuel's (RCF or RFNBO) molecular structure.

#### D.9.2.2. Electricity Inputs

For electricity inputs that enhance the heating value of the fuel or intermediate products, the relevant energy is the energy of the electricity. If the LHV of the output fuel exceeds that of the inputs, then the electricity inputs are adding to the fuel's heating value and must be treated as relevant energy for emissions calculations.

#### D.9.2.3. Heat Inputs

For heat inputs, the relevant energy is the useful energy in the heat used to synthesize the fuel. Useful heat is calculated as the total heat energy multiplied by the Carnot efficiency (refer to Section D.4.). If the LHV of the output fuel exceeds that of the inputs, then the heat inputs are adding to the fuel's heating value and must be treated as relevant energy for emissions calculations.

#### D.9.2.4. Industrial Off-Gases

For industrial off-gases, the relevant energy is the energy in the off-gas based on the respective LHV. In the case of heat that is used to enhance the heating value of the fuel or intermediate product, the relevant energy is the useful energy in the heat used to synthesize the fuel. Useful heat is the total heat energy multiplied by the Carnot efficiency[13].

### D.9.3. Emissions Intensity

Upon determination of the output share of RCF or RFNBO (refer to Section D.9.1.) and the relevant energy inputs (refer to Section D.9.2.), the emissions intensity shall be determined on a proportional basis of the energetic value of inputs between:

- The part of the process that is based on the conventional input, and
- The part of the process that is based on the RCFs and RFNBOs assuming that the process parts are otherwise identical.

[13] Refer to Part C Point (1) (b) Annex V of the *Revised Directive (EU) 2018/2001 (RED III)*.

The energetic share of all inputs must be respected for the virtually split part. Emissions intensity for the RCF or RFNBO fraction of the total fuel output shall then be determined using the following equation.

$$Em_{RFNBO/RCF}\left[\frac{g\ CO_{2eq}}{MJ\ fuel}\right] = \sum_{n}(S_{RFNBO/RCF} \times e_{i,n}) + e_p + e_{td} + e_u - e_{CCS}$$

Where:

$Em_{RFNBO/RCF}$: Total emissions assigned to RFNBO or RCF.

*n:* Each "relevant" input.

$e_{i,n}$: Emissions intensity for the supply of the input "n".

$e_p$: Emissions from processing.

$e_{td}$: Emissions from transport and distribution.

$e_u$: Emissions from combusting the fuel in its end use.

$e_{ccs}$: Emission savings from carbon capture and geological storage.

## D.10. Flexible Allocation

Participating operators can choose to apply the mechanism of flexible allocation. When a share of the fuel output is RCF or RFNBO, the participating operator can proportionally attribute emissions from processing inputs or feedstocks to specific products or outputs to adjust their emission reduction efforts.

Refer to Section D.9.2. to determine the relevant energy inputs and Section A.6.2. for eligible RCF and RFNBO feedstocks/inputs.

# ANNEXES

## Annex 1: History of changes

*Table 4: History of Changes to the RSB GHG Methodology documentation.*

| Version/Date | Section(s) | Change |
|---|---|---|
| 10.09.12 | 1.3 | Added section 1.3: GHG Emissions calculation through the chain of production |
| 10.09.12 | 1.4 | Expanded the section on system boundary providing a more detailed explanation of all the inputs that need to be taken into consideration and introducing the concept of materiality threshold. |
| V2.2 / 03.05.16 | Throughout the document | Included requirements of the European Commission for full compliance under the EU Renewable Energy Directive. |
| V2.3 / 08.08.17 | 3.2.4 | The baseline date for land use change emissions was changed to January 2008 (consistent with the revision of the RSB Principles & Criteria). |
| V3.0 / 13.06.25 | Throughout the document | Aligned the emission factors for the RSB methodology with EU RED.<br>Reporting GHG assessment results changed from kilogram to gram $CO_{2eq}$ for both fuel and bio-based products.<br>Combined Heat and Power plants emissions calculations are aligned to EU RED.<br>Calculations for nitrous oxide emissions as part of land cultivation operations are aligned to EU RED.<br>Land use change emission calculations are aligned to IPCC's 2019 Refinement to the 2006 Guidelines for National Greenhouse Gas Inventories Volume 4.<br>Co-processing of biofuels, RCFs and RFNBOs are incorporated alongside flexible allocation. |

## Annex 2: Eligible Biofuel and Biomaterial Feedstocks

*Table 5: Biofuel and Biomaterial feedstock crops considered eligible under RSB GLOBAL.*

| Eligible Feedstock Crops |
|---|
| Barley |
| Camelina |
| Cassava |
| Castor bean |
| Coconut |
| Corn |
| Eucalyptus cuttings |
| Jatropha |
| Maize |
| Palm |
| Poplar cuttings |
| Potato |
| Rapeseed |
| Residue – Agricultural |
| Residue – Forestry |
| Rye |
| Soybean |
| Sugar beet |
| Sugarcane |
| Sunflower |
| Sweet sorghum |
| Wheat |

Note: Refer to *RSB Standard for Advanced Fuels (RSB-STD-01-010)* and *RSB EU RED Standard for Advanced Fuels (waste and residues) (RSB-STD-11-001-01-010)* for detailed feedstock-specific requirements.

## Annex 3: Lower Heating Values

*Table 6: List of Lower Heating Values.*

| Fuel Type | Lower Heating Value [MJ/kg-dry basis] |
|---|---|
| Animal fat | 38.8 |
| Bagasse | 17.0 |
| Bio-Gasoline | 44.0 |
| Bio-Oil (byproduct FAME from waste oil) | 21.8 |
| Biowaste | 20.7 |
| Corn (maize grain only) | 17.3 |
| Crude vegetable oil | 37.0 |
| DDGS (maize) | 19.2 |
| Diesel | 43.1 |
| Ethanol | 26.8 |
| Eucalyptus | 19.0 |
| FAME | 37.2 |
| Fatty acids | 37.0 |
| FFB | 24.0 |
| Forestry residues | 19.0 |
| Gasoline/Petrol | 43.2 |
| Glycerol | 16.0 |
| Hard coal | 26.5 |
| HVO | 44.0 |
| Hydrogen (from renewable sources) | 120.0 |
| Lignite | 9.2 |
| LPG | 46.2 |
| Manure | 12.2 |
| Methanol | 19.9 |
| n-Hexane | 45.1 |
| Natural Gas | 49.2 |
| Palm kernel meal | 18.5 |
| Palm oil | 37.0 |
| PVO | 36.0 |
| Rapeseed | 27.0 |
| Rapeseed oil cake | 18.4 |
| Sawdust | 19.0 |
| Soybean oil cake | 19.1 |
| Soybeans | 23.0 |
| Straw | 17.2 |
| Sugar beet | 16.3 |
| Sugar beet pulp | 16.1 |
| Sugar cane | 19.6 |
| Sunflower seed | 27.2 |

| | |
|---|---|
| Syn diesel (BtL) | 44.0 |
| Vinasse | 37.0 |
| Waste cooking oil | 37.0 |
| Wheat | 17.0 |
| Wheat straw | 17.2 |

## Annex 4: References


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3. Commission Implementing Regulation (EU) 2022/996 of 14 June 2022 (https://eur-lex.europa.eu/eli/reg_impl/2022/996/oj/eng).
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17. RSB Standard for Advanced Fuels (RSB-STD-01-010; https://rsb.org/wp-content/uploads/2024/06/RSB-STD-01-010-RSB-Standard-for-advanced-fuels.pdf).
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